

Customer Churn

Analysis & Prediction

A data-driven business case for understanding, predicting, and reducing customer attrition using the IBM Telco Customer Churn Dataset.

Dataset	IBM Telco Churn (Kaggle / IBM Cognos)
Domain	Telecommunications
Records	7,043 Customers · 21 Features
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26.5%	7,043	\$65.05	~80%	+17pp
Overall Churn Rate	Total Customers	Avg Monthly Charge	Model Accuracy	Recall Gain

1. Business Context

Customer churn — the rate at which subscribers cancel services — is one of the most significant levers affecting revenue in the telecommunications industry. Acquiring a new customer costs five to seven times more than retaining an existing one, making churn prevention a high-return investment for any telecom operator.

The IBM Telco Customer Churn Dataset models a fictional telecom company facing a churn rate of **26.5%** — a level that materially erodes subscriber lifetime value and long-term revenue stability. Without a systematic, data-driven approach, the business cannot proactively identify at-risk customers or deploy retention resources efficiently.

2. Problem Statement

Lack of Churn Visibility

The business has no systematic way to identify which customers are likely to leave before they do.

	Without EDA and statistical analysis, the drivers of churn remain invisible — preventing targeted intervention.
	Current strategies are reactive (responding after cancellation) rather than proactive, resulting in higher cost-per-save.
	Without a risk score, retention budget is spread evenly across all customers rather than concentrated on high-risk segments.

3. Project Objectives

01	Identify Key Churn Drivers Perform EDA to surface the contract types, pricing bands, and service usage patterns most strongly correlated with churn.
02	Build a Predictive Model Train a Logistic Regression classifier to score each customer's churn probability with ~80% accuracy.
03	Optimise Detection Recall Lower the classification threshold from 0.5 to 0.3 to improve churn recall from ~57% to ~74%, prioritising early identification.
04	Segment & Prioritise Risk Classify customers into High / Medium / Low risk tiers to enable tiered, cost-effective retention campaigns.
05	Enable Scalable Deployment Serialise the trained model (joblib) for reuse without retraining, and deliver a Power BI dashboard for business stakeholders.

4. Key Findings from EDA

Finding	Churn Signal	Business Implication
Month-to-month contracts	↑ HIGHEST	Highest churn cohort; long-term plans urgently needed
Tenure < 3 months	↑ HIGH	Onboarding failure is the primary early-life risk

Monthly charges > \$65	↑ HIGH	Price sensitivity peaks in mid–high tier customers
No Tech Support subscription	↑ MODERATE	Low engagement signals disengagement and churn intent
Electronic check payment	↑ MODERATE	Correlates with lower commitment vs auto-pay users
Two-year contracts	↓ LOWEST	Strong retention anchor; upsell path for at-risk users

5. Customer Risk Segmentation & Retention Playbook

■ HIGH RISK	Profile: • Tenure < 3 months • Month-to-month contract • High charges, low engagement	Actions: • Launch targeted onboarding campaigns • Offer discounted annual/biannual plans • Assign dedicated account manager touch
■ MEDIUM RISK	Profile: • Moderate tenure (3–12 months) • Partial service adoption • No tech support usage	Actions: • Send feature discovery nudges • Recommend tech support bundles • Mid-contract loyalty incentives
■ LOW RISK	Profile: • Tenure > 12 months • Auto-payment enrolled • Multi-service subscriber	Actions: • Enrol in loyalty reward programmes • Cross-sell premium tiers • Annual satisfaction check-ins

6. Predictive Model & Performance

Metric	Default Threshold (0.5)	Optimised Threshold (0.3)
Overall Accuracy	~80%	~80%
Churn Recall	~57%	~74%
False Positive Rate	Lower	Higher
Business Use	General scoring	Retention-focused targeting

A threshold of 0.3 is recommended for production deployment. While it increases false positives, the cost of a missed churner (lost CLV) significantly outweighs the cost of a false-positive retention touchpoint.

7. Business Impact & Future Roadmap

Immediate Impact	• Churn detection improved from ~57% to ~74% recall • Early identification of at-risk customers before cancellation • Proactive retention replaces reactive save-desk cost
Future Roadmap	• Upgrade to Random Forest / XGBoost for higher accuracy • Build a real-time REST API for live churn scoring • Connect Power BI dashboard to production database